(11) Charge the refrigerant system (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - STANDARD PROCEDURE - REFRIGERANT SYSTEM CHARGE).

A / C CONDENSER

DESCRIPTION

The A/C condenser is located in the front of the engine compartment behind the grille (Fig. 7). The A/C condenser is a heat exchanger that allows the high-pressure refrigerant gas being discharged by the A/C compressor to give up its heat to the air passing over the condenser fins, which causes the refrigerant to cool and change to a liquid state.

The A/C condenser is equipped with mounting provisions and a tapping block for the A/C refrigerant lines.

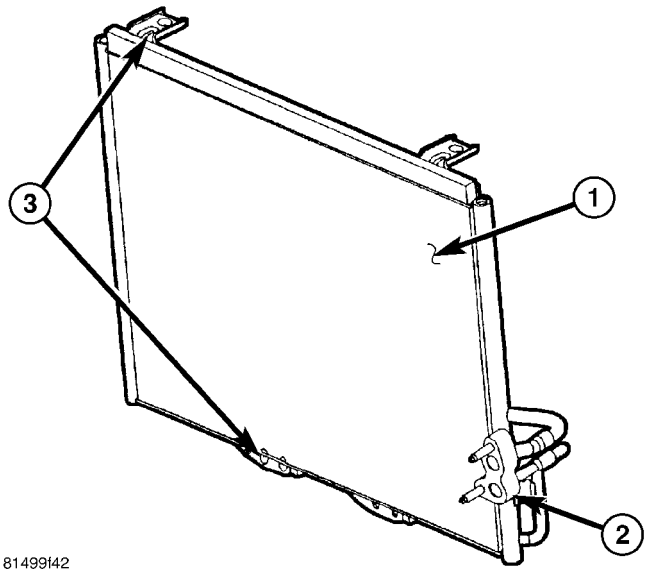

Fig. 7 A/C Condenser - Typical

1 - A/C CONDENSER
2 - MOUNTING BRACKET
3 - REFRIGERANT LINE TAPPING BLOCK

OPERATION

When air passes through the fins of the A/C condenser, the high-pressure refrigerant gas within the A/C condenser gives up its heat. The refrigerant then condenses as it leaves the A/C condenser and becomes a high-pressure liquid. The volume of air flowing over the condenser fins is critical to the proper cooling performance of the A/C system. Therefore, it is important that there are no objects placed in front of the radiator grille openings at the front of the vehicle or foreign material on the condenser fins that might obstruct proper air flow. Also, any factory-installed air seals or shrouds must be properly reinstalled following radiator or A/C condenser service.

The A/C condenser cannot be repaired and, if faulty or damaged, it must be replaced.

REMOVAL

WARNING: Refer to the applicable warnings and cautions for this system before performing the following operation (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - WARNINGS) and (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - CAUTIONS). Failure to follow the warnings and cautions could result in possible personal injury or death.

CAUTION: Before removing the A/C condenser, note the location of each of the radiator/condenser air seals. These air seals are used to direct air through the A/C condenser and radiator. The air seals must be reinstalled in their proper locations in order for the A/C and engine cooling systems to perform as designed.

(1) Disconnect and isolate the negative battery cable.

(2) Recover the refrigerant from the refrigerant system (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - STANDARD PROCEDURE - REFRIGERANT SYSTEM RECOVERY).

(3) Partially drain the engine cooling system (Refer to 7 - COOLING - STANDARD PROCEDURE - COOLING SYSTEM DRAINING).

(4) Remove the cooling recovery container (Refer to 7 - COOLING/ENGINE/COOLANT RECOVERY CONTAINER - REMOVAL.

(5) Disconnect the upper radiator hose from the radiator and position the hose out of the way (Refer to 7 - COOLING - OPERATION - HOSE CLAMPS).

(6) Remove the radiator cooling fan and fan shroud (Refer to 7 - COOLING/ENGINE/FAN DRIVE VISCOUS CLUTCH - REMOVAL).

(7) Raise and support the vehicle.

(8) Position the lower condenser/radiator air seal out of the way and remove the two bolts that secure the lower condenser mounting bracket to the bottom of the grille panel.

(9) Lower the vehicle.

(10) If equipped, remove the upper condenser/radiator air seal (Fig. 8).

(11) Remove the nuts that secure the discharge and liquid lines to the condenser tapping block.

(12) Disconnect the discharge and liquid lines from the condenser tapping block and remove and discard the O-ring seals.

(13) Install plugs in, or tape over the opened refrigerant line fittings and the condenser ports.